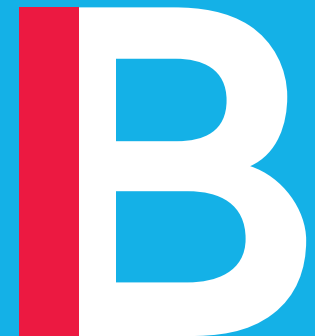


Quantitative Methods

Instrumental Variables

by Professor Shuang Zhang



Housekeeping



- Problem set 1 is posted on the Hub
- It is only for your practice; no submission required
- Questions have similar format to the final exam
- Answers will be posted after the presentation week

Recap: regression and CIA

$$Y_i = \alpha + \beta D_i + \gamma X_i + \varepsilon_i$$

- Conditional Independence Assumption
- ***Other Things Equal***: any unobservable (cannot be included in X) that could violate this principle?
- Other non-experimental paths to *Other Things Equal*
 - **Instrumental Variable**
 - Regression Discontinuity Design
 - Difference-in-Difference

Do *better* public schools matter?

Charter schools in the US

Charter schools in the US

- Many children in large urban districts leave school with poor reading and math skills.
- Charter schools—privately managed public schools—offer a possible solution
 - Charters are more autonomy in teaching hiring, curricular structure and school hours.
 - The Knowledge is Power Program (KIPP) is serving mostly urban minority students.
 - KIPP students tend to do better than other students in the district. But is this selection bias or a causal effect?

Charter schools in the US

Here's what the critics say:

KIPP students, as a group, enter KIPP with substantially higher achievement than the typical achievement of schools from which they came. . . . Teachers told us either that they referred students who were more able than their peers, or that the most motivated and educationally sophisticated parents were those likely to take the initiative . . . and enroll in KIPP.

Selection bias

- Let D_i denote attendance at KIPP and Y_i be an achievement test outcome
- Under constant causal effects,

$$Y_i = Y_{0i} + (Y_{1i} - Y_{0i})D_i = \alpha + \lambda D_i + \eta_i, \quad (4)$$

where λ is yet another Greek name for the causal effect of interest. Again, we're confounded by selection bias:

$$E[Y_i|D_i = 1] - E[Y_i|D_i = 0] = \lambda + \{E[\eta_i|D_i = 1] - E[\eta_i|D_i = 0]\}$$

Define instrumental variable (IV)

- An instrument (Z_i) for KIPP attendance in (4) is correlated with D_i but uncorrelated with Y_{0i} . **In the context of equation (4), instrumental variable Z_i is assumed to satisfy:**

$$C(Z_i, D_i) \neq 0 \tag{5}$$

$$C(Z_i, \eta_i) = 0 \tag{6}$$

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$$C(Z_i, D_i) \neq 0 \quad (5)$$

$$C(Z_i, \eta_i) = 0 \quad (6)$$

If both (5) and (6) are satisfied, you run two regressions separately

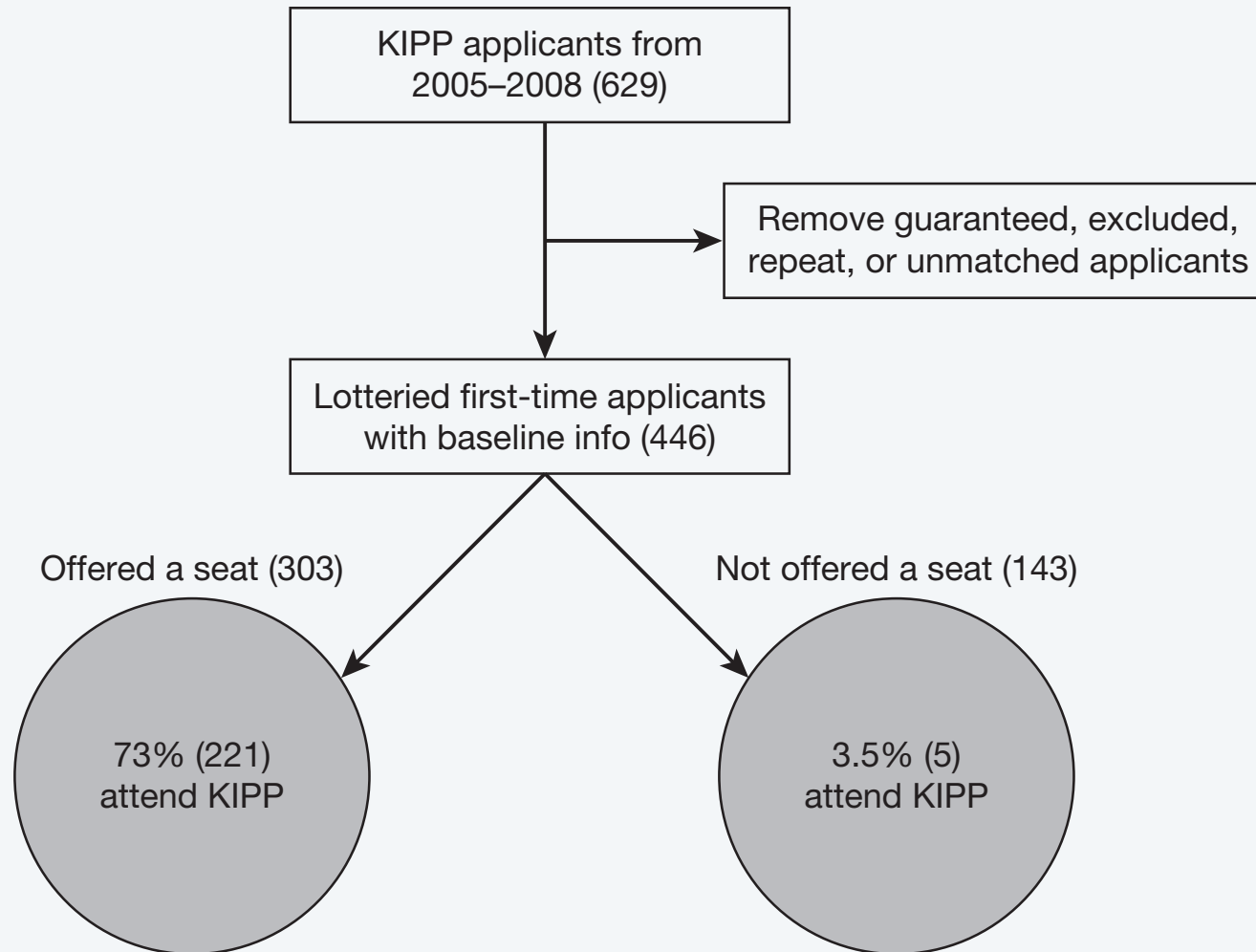
- $Y_i = \alpha + \rho Z_i + \varepsilon_i$ (The reduced form)
- $D_i = \alpha + \phi Z_i + \mu_i$ (The first stage)

$$\frac{\text{The Reduced Form}}{\text{The First Stage}} = \frac{\rho}{\phi} = \lambda_{IV}$$

Application: the KIPP lottery

- Like all Massachusetts charter schools, KIPP Lynn assigns seats by lottery (i.e., at random) when over-subscribed
 - A research jackpot!
- In this case, instrument Z_i is a dummy variable indicating the set of KIPP applicants randomly offered a KIPP seat
 - Because the lottery is how most KIPP applicants get seated there, (5) is satisfied
 - Because lottery offers are randomly assigned, they're likely to be independent of potential outcomes, satisfying (6)

FIGURE 3.1
Application and enrollment data from KIPP Lynn lotteries



Note: Numbers of Knowledge Is Power Program (KIPP) applicants are shown in parentheses.

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Random assignment balances pre-existing characteristics

	KIPP applicants				
	Lynn public fifth graders (1)	KIPP Lynn lottery winners (2)	Winners vs. losers (3)	Attended KIPP (4)	Attended KIPP vs. others (5)
Panel A. Baseline characteristics					
Hispanic	.418	.510	−.058 (.058)	.539	.012 (.054)
Black	.173	.257	.026 (.047)	.240	−.001 (.043)
Female	.480	.494	−.008 (.059)	.495	−.009 (.055)
Free/Reduced price lunch	.770	.814	−.032 (.046)	.828	.011 (.042)
Baseline (4th grade) math score	−.307	−.290	.102 (.120)	−.289	.069 (.109)
Baseline (4th grade) verbal score	−.356	−.386	.063 (.125)	−.368	.088 (.114)
Panel B. Outcomes					
Attended KIPP	.000	.787	.741 (.037)	1.000	1.000 —
Math score	−.363	−.003	.355 (.115)	.095	.467 (.103)
Verbal score	−.417	−.262	.113 (.122)	−.211	.211 (.109)

IV estimator

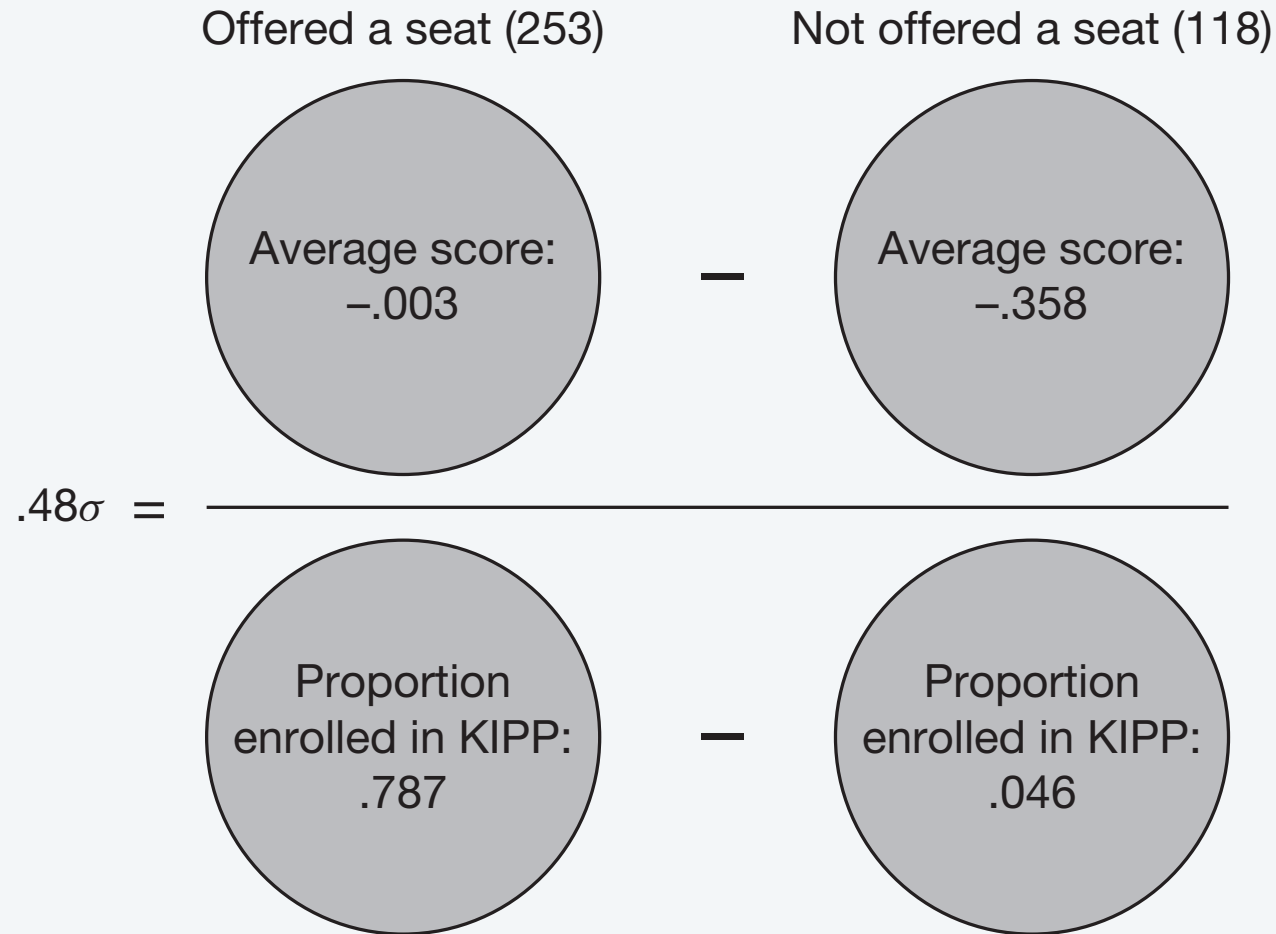
$$\lambda_{IV} = \frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{E[D_i|Z_i = 1] - E[D_i|Z_i = 0]}$$

IV estimator

$$\lambda_{IV} = \frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{E[D_i|Z_i = 1] - E[D_i|Z_i = 0]}$$

- First stage: $E[D_i|Z_i = 1] - E[D_i|Z_i = 0] = \phi$
 - Difference in KIPP attendance rates between those who were and were not offered a seat in the lottery
- Reduced form: $E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0] = \rho$
 - Difference in average test scores between those who were and were not offered a seat in the lottery
- $\lambda_{IV} = \rho/\phi$
- IV estimator is also called 2SLS (two stage least squares)

FIGURE 3.2
IV in school: the effect of KIPP attendance on math scores



Note: The effect of Knowledge Is Power Program (KIPP) enrollment described by this figure is $.48\sigma = .355\sigma / .741$.

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Different types of children

- KIPP lottery offers affect KIPP enrollment for many applicants . . . but not all
 - Some offered a seat at KIPP nevertheless go elsewhere
 - A few not offered a seat in the lottery manage to sneak in anyway
- How should we interpret IV estimates in light of this fact?

TABLE 3.2
The four types of children

		Lottery losers $Z_i = 0$	
		Doesn't attend KIPP $D_i = 0$	Attends KIPP $D_i = 1$
Lottery winners $Z_i = 1$	Doesn't attend KIPP $D_i = 0$	Never-takers (<i>Normando</i>)	Defiers
	Attends KIPP $D_i = 1$	Compliers (<i>Camila</i>)	Always-takers (<i>Alvaro</i>)

Note: KIPP = Knowledge Is Power Program.

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LATE: who benefits from Charter schools?

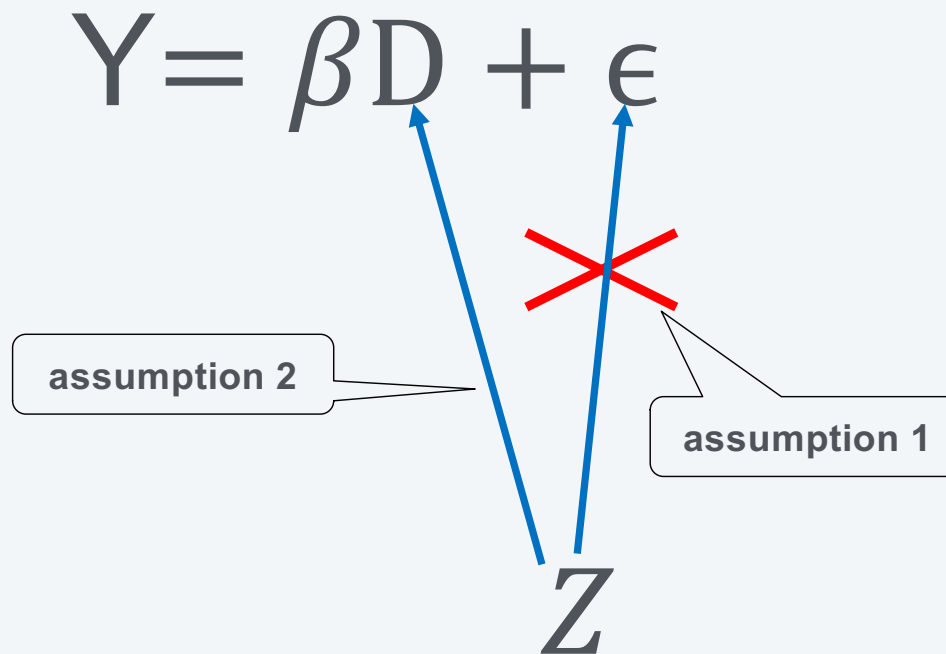
- In a world of heterogeneous potential outcomes,

$$\lambda_{IV} = \frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{E[D_i|Z_i = 1] - E[D_i|Z_i = 0]} = E[Y_{1i} - Y_{0i}|C_i = 1],$$

where C_i indicates compliers, like Camila

- Parameter $E[Y_{1i} - Y_{0i}|C_i = 1]$ is called a *local average treatment effect* (LATE)
- **LATE**: average treatment effect for compliers - children whose KIPP enrollment status is determined solely by the KIPP lottery

A graphical representation of IV



Two assumptions for valid instrumental variables

1. Independence assumption: independent of error term
2. There is a first stage: affect variable of interest

Must be argued

Can be checked

Non-experimental IV: Example 1)

- Are large families good or bad for children's education outcome?
 - Quantity-quality tradeoff: family size and living standards
 - But cannot run lottery on family size!



Family size and children's education outcomes

(Study by Angrist, Lavy & Schlosser)

There is some randomness in family size (independence assumption)

Two factors that are out of control of controlling parents:

- Occurrence of twins
- Sex of baby

Ok those might have a reasonable independence assumption, but does it have a first stage?

- Twins: families might only have planned for 2 kids, but when they had twins they un-intentionally had 3
- Many families have preference for a sex mix (a boy and a girl)
- Hence, if they have two kids of the same sex they are more likely to carry on having more kids

First stage: IV and family size

TABLE 3.4
Quantity-quality first stages

	Twins instruments		Same-sex instruments		Twins and same- sex instruments
	(1)	(2)	(3)	(4)	(5)
Second-born twins	.320 (.052)	.437 (.050)			.449 (.050)
Same-sex sibships			.079 (.012)	.073 (.010)	.076 (.010)
Male		-.018 (.010)		-.020 (.010)	-.020 (.010)
Controls	No	Yes	No	Yes	Yes

2SLS: family size and children's education outcomes

TABLE 3.5
OLS and 2SLS estimates of the quantity-quality trade-off

Dependent variable	OLS estimates (1)	2SLS estimates		
		Twins instruments (2)	Same-sex instruments (3)	Twins and same- sex instruments (4)
Years of schooling	−.145 (.005)	.174 (.166)	.318 (.210)	.237 (.128)
High school graduate	−.029 (.001)	.030 (.028)	.001 (.033)	.017 (.021)
Some college (for age ≥ 24)	−.023 (.001)	.017 (.052)	.078 (.054)	.048 (.037)
College graduate (for age ≥ 24)	−.015 (.001)	−.021 (.045)	.125 (.053)	.052 (.032)



Non- experimental IV: Example 2)

What are the impact of air pollution on health?

- Economic activities can increase pollution
- Economic activities increase income and improve health
- Ignoring economic factors could lead to an upward bias by creating a spurious positive correlation between pollution and health

Airports, Air pollution and Health (Study by Schlenker and Walker, 2016)

There is also some randomness in air pollution (independence assumption)

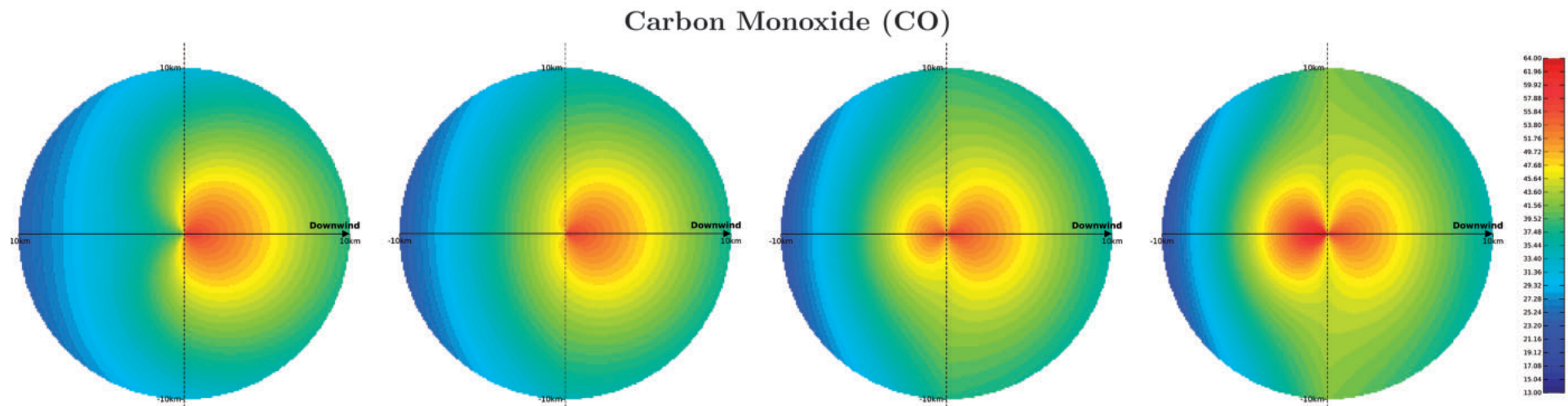
- Excess airplane idling (measured as residual daily taxi time) is due to network delays

Ok those might have a reasonable independence assumption, but does it have a first stage?

- Airplane taxi time: does it affect local air pollution?
- Hypothesis: it could increase carbon monoxide (CO) exposure depending on wind direction and speed

First stage: IV and local air pollution

- CO levels in neighbourhoods near the airport: both upwind and downwind directions
- From left to right below: wind speed from weak to strong



2SLS: air pollution and health

- Correlation shows almost no impact of pollution in health
- IV estimators below: positive impacts on pollution-related health outcomes

	Asthma (1a)	Acute respiratory (1b)	All respiratory (1c)	Heart problems (2)
Panel A: all ages				
Model 1: CO	0.311*** (0.065)	0.556*** (0.162)	0.761*** (0.207)	0.434*** (0.134)

Limitations of IV method

- LATE: internal validity
 - Effects on compliers
 - Doesn't say anything about non-compliers
- External validity?
 - When treatment status is not determined through the IV mechanism, IV estimators may not apply

ANY
QUESTIONS
?